Williams College
SMALL Undergraduate Research Program
Application Form

You should apply online at http://www.mathprograms.org/db (below is the additional info required; you can type directly on this sheet or print it out and upload a scan). Remember to apply only to your top choice; if you apply to more your application will not be read.

Full Name: Niyathi Kukkapalli

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College or University: Princeton University Expected Graduation Date: May 2028

Write Current Status: first-year sophomore junior other (explain)

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How did you hear about SMALL? Through PROMYS

You may rank up to three groups in which you are interested below. For updates and descriptions see http://math.williams.edu/small/ or the MathJobs page. Your first choice should be the one you are applying to.

First choice: Chip-firing Games on Graphs

Second choice: Number Theory and Probability

Third choice: Optimal Control Problems in Binocular Vision

List of math courses completed or currently being taken, with grades for those completed (if this is not enough room just add additional lines).

Honors Linear Algebra

Mathematical Game Theory

Single Variable Analysis & Intro to Proofs (B)

Also have at least one relevant faculty member upload a letter of recommendation by the deadline. (For Williams students, just give us at least one name of a math faculty member who can serve as a reference.) Please list the name and email address of your reference here.

Prof. Li Mei-Lim, mei121@bu.edu

US citizenship status. (Funding is available for a limited number of students who are not US citizens or permanent residents.)

✓ U.S. Citizen ________Permanent Resident _______Non-U.S. Citizen

Statement of interest (1-2 pages): *(Attach at the end of this form)*

All application materials must be received on-line by **the listed date for full consideration (see the webpage, usually the first or second Wednesday in February)**; if you hear from another program that has an earlier deadline than ours but are interested in SMALL, please let us know so we can give you an update on your status.

Questions: See the FAQ at https://math.williams.edu/small-application-faq/ or email the SMALL Director at smalldirector@williams.edu.

It’s in the mail.
It’s here!
I rip open the package.

Opening the comic book, sniffing the pages redolent of ink fresh from the factory, I gloss over the cover, *The Art of Problem Solving*. It’s like Harry Potter–but instead of Harry and Hogwarts, there’s Beasts and Math Competitions! I was dazzled by the flying dragon who could multiply, the pirate seal who could calculate areas, and the 8-legged camel who could count. Later I branched out from *AoPS* comics to *AoPS* textbooks, I spent hours everyday solving math problems. The process of learning and applying theorems from *Ptolemy* to *Brahmagupta,* was my first glimpse of mathematics.

While these books fueled my love for problem-solving, I still didn’t fully grasp the logic behind the theorems—the deeper *why* behind the answers. So I set out to find it by spending my last two summers of high school at PROMYS (Program in Mathematics for Young Scientists) doing exactly that. But it was only when one of my problem sets asked me, “Prove A x 0 = 0” that I truly understood the beauty of proofs or the why behind an answer. A x 0 = 0? *Well, isn’t that supposed to be obvious?* That’s what I thought too. But actually, dozens of inverse, distribution, and addition axioms are used to prove it. *Isn’t that so cool?* It clicked–the crux of mathematics is actually *proving* why your answer is correct.

In my early highschool years, a typical evening was me at my desk, with papers sprawled out, working on a math problem in solitude. But my past two summers at PROMYS have given me something truly indispensable. From everyone collectively bashing our heads against our problem-sets to blasting youtube karaoke–I found a community. A community of math-geeks. A community of hilarious people. And a community where I learned what collaboration in mathematics truly meant. But when I left, I found myself with even more questions: What would it be like to do real research? To work on open problems rather than structured coursework? To apply mathematical ideas beyond Number Theory?

I *tried* to explore these questions. At the Australian National University this past winter, I analyzed upper bounds for integrals of periodic solutions to the Schrödinger equation, using Mathematica and Python to visualize exponential sums. At the University of Delaware, I worked on establishing a lower boundary for disconnecting vertex sets in Hamming graphs, diving into Spectral Graph Theory to improve known bounds. At PROMYS, on the side of my Number Theory coursework, I studied p-adic numbers and proved properties of the Artin-Hasse exponential. And at Princeton, I find myself attending the Math Department’s Colloquium on Topology or attempting the extra credit on our Analysis problem set–but something still feels missing.

While I’ve explored research in different settings, it’s always felt somewhat fragmented—independent, unstructured, and squeezed between other commitments. Each experience left me with more curiosity but also more uncertainty: Do I truly enjoy research, or have I just never had the chance to fully immerse myself in it? What I've been missing is a research environment with real structure—one where I can collaborate with mentors and peers, focus entirely on the problem at hand, and see a project through with the support and resources to get meaningful results. At SMALL I can picture myself working[on an open problem with guidance, learning how to navigate the unknown, and contributing something new to the field. I want to immerse myself in a research community where I can explore mathematics beyond what I know and see firsthand how abstract ideas evolve into discoveries.

I'm a college freshman, so I don't really know what the future holds. While I have my heart set on becoming a mathematician and contributing to the corpus of proofs that make up mathematics, I can also picture myself using math to solve real-world problems perhaps in industry. At SMALL, by 12 o'clock I can study my research background review and by 5 o'clock I'm coming back from a group talk of recent papers in mathematics and going to play ultimate frisbee–the balance of focused academic work and diverse social engagement will make my immersion into research even more fruitful. Even if a lot is uncertain, I'm sure William's REU's can push me one step forward in the right direction–one proof at a time.